

PROCESS OR MATERIALS REVIEW REQUEST FORM

Request Title: SLAC CDMS Ge Fabrication

Requestor: Matthew Cherry

Date: 1/28/20

Non-SNF staff, please complete the following:

Badger ID: MCherry

Company name/PI: SLAC

Send completed form to: snf-promcommittee@lists.stanford.edu

1. **The chemical or material.** Please provide all common names, trade names, and CAS numbers where appropriate. Include an MSDS, if available; or provide the reason, if not. Make sure to include information for any new secondary chemicals (such as a developer for a new resist). Read the MSDSs as well as the [Stanford Chemical Storage Groups](#) to determine the Storage Group Identifier and Main Hazard Class of your chemical/material.

Germanium substrates with sputtered layers of silicon, tungsten, and aluminum. All depositions done in a dedicated, clean machine in PAB on the Stanford campus.

2. **Vendor/manufacture info:** address and phone number, website URL.

Germanium was acquired from Umicore, in Belgium. I can provide more detail regarding this if needed. The sputtering targets are acquired from VEM (Si & W) and AJA, Inc. (Al).

3. **Reason for request:** Please give serious thought to this. If you have any process information (application notes from the vendor, protocol from another lab, experimental methods section of an article), please include it, preferably as attachments. Ask yourself these questions: Is this the latest procedure? Are there newer/safer alternatives that will also work for my project? Will any of the current SNF approved chemicals and materials work for me?

CDMS has been using SNF for our fabrication process for 20+ years. The recipe is mature and actually currently in use. This request came up from a discussion between Matt Cherry and Uli regarding the (potential) new wbmatal and the somewhat-recently-installed germanium wetbench.

4. **Process Flow:** Please provide a detailed process flow description on how and where you proposed to use this chemical. This should include **all Lab equipment** to be used for processing your wafers once your new chemical or material has been used (even if your new material is a film that is removed, it may still pose potential contamination concerns.)

Make sure to include wet benches. Please note that if the chemical/material is to be used in any the "clean" equipment, purity specifications will be needed. This is most important for chemicals/material that are not normally used for VLSI device fabrication. To be allowed into a "clean" tool, the material should be MOS grade or better.

The existing recipe is as follows:

All work done in dedicated cassettes.

1. *Germanium polished to semiconductor spec in an underground facility on the Stanford campus.*
2. *Solvent clean at Lithography Solvent bench. Xylenes 15', Acetone 5', IPA 5'.*
3. *9% HF dip for 8' at wbmatal/wbclean3.*
4. *Offsite deposition of silicon and tungsten layers.*
5. *Photoresist coating done at CDMS Laurel Spinner, (adjacent to Litho Solvent), S1813.*
6. *Pre-bake done in 90C oven.*
7. *Exposure done on evalign2.*
8. *Developing done with MF319 in dedicated glassware at litho developer bench.*
9. *Post-bake done in 110C oven.*
10. ***Tungsten layer etched in Hydrogen Peroxide (H₂O₂) for ~400s at wbmatal/wbclean3.***
11. *Photoresist strip with SRS100 10' & PRS1000 10' at wbmatal/wbclean3.*
12. *Offsite deposition of aluminum layer.*
13. *Photoresist coating done at CDMS Laurel Spinner, (adjacent to Litho Solvent), S1813.*
14. *Pre-bake done in 90C oven.*
15. *Exposure done on evalign2.*
16. *Developing done with MF319 in dedicated glassware at litho developer bench.*
17. *Post-bake done in 110C oven.*
18. ***Aluminum layer etched in Aluminum etchant in 45s increments with 15s DI water quench, done at wbmatal/wbclean3.***
19. *Photoresist strip with SRS100 10' & PRS1000 10' at wbmatal/wbclean3.*
20. *Silicon layer processing steps below:*
21. *Photoresist coating done at CDMS Laurel Spinner, (adjacent to Litho Solvent), S1813.*
22. *Pre-bake done in 90C oven.*
23. *Exposure done on evalign2.*
24. *Developing done with MF319 in dedicated glassware at litho developer bench.*

25. Post-bake done in 110C oven.

26. Silicon layer etched in the Oxford RIE, 100W, 150mT, 40sccm CHF3, 20sccm O2, 7'.

27. Photoresist strip with SRS100 10' & PRS1000 10' at wbmatal/wbclean3.

5. **Amount and form.** How much will you bring in? Is it solid, powder or liquid? (Note: as a general rule, powders are not permitted in the cleanroom.) Do you need to mix it to use it?

The only outside chemical used in this process is the photoresist, Shipley S1813. We use 1qt. bottles.

6. **Storage:** Will you be storing your chemical/material at SNF? If so, please note any potential reactivity's (this should be on the MSDS). [Storage groups](#) A, B, D and L are stored in the yellow solvent cabinet in the furnace support area, while [storage groups](#) C, E, F and G are stored on top of one of the Pass-through Carts. Ensure your chemical container or material is properly labeled. If there is no available room, it must be stored by in the bulk storage area. You will then need to obtain it from receiving area personnel each time you want to use it and return it to them when you are finished using it (or each time you leave the lab). Note that there is no storage of chemicals/materials in the processing lab or at any wet bench.

S1813 is in storage group L.

7. **Disposal:** How will you dispose of any waste or excess chemical or material? In your discussions with experts and vendors, try to determine the best way to dispose of your spent chemicals and by-products. Please refer to the [SNF Lab members Safety Manual](#) for the different methods of waste disposal that are available in the lab.

Nothing special is done for our process, as all chemical (aside from the photoresist) are supplied by SNF. Photoresist has been captured at the CDMS Laurel and disposed of by Mario once/year.

PROCESS OR MATERIALS REVIEW REQUEST FORM

To be completed by the PROM COMMITTEE:

Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date: